Mansoura University
Faculty of Computers and Information
Department of Information System
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[IS312P] System Analysis and logical design

Grade: 3th IS

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Lecture 6
Introduction to Requirement Modeling
Agile Age

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Agile Age

Agile Methods

- Agile Approaches
  - Scrum
  - Kanban
- Agile pros and cons

Modeling Tools and Techniques

- FDD
- BPMN
- DFD
- UML

Agile Methods

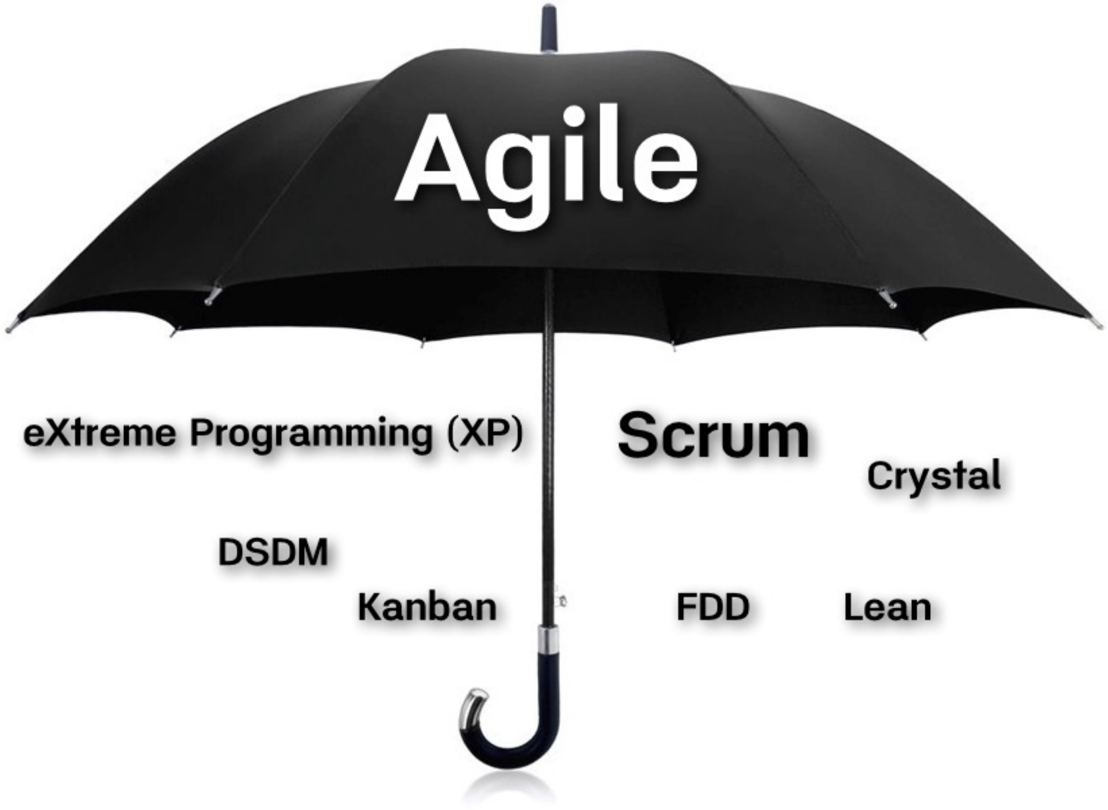
Agile methods attempt to develop a system incrementally, by building a series of prototypes and constantly adjusting them to user requirements.

Agile

An agile approach emphasizes continuous feedback, and each incremental step is affected by what was learned in the prior steps.

Many agile developers prefer not to use CASE tools at all, and rely instead on whiteboard displays and arrangements of movable sticky notes

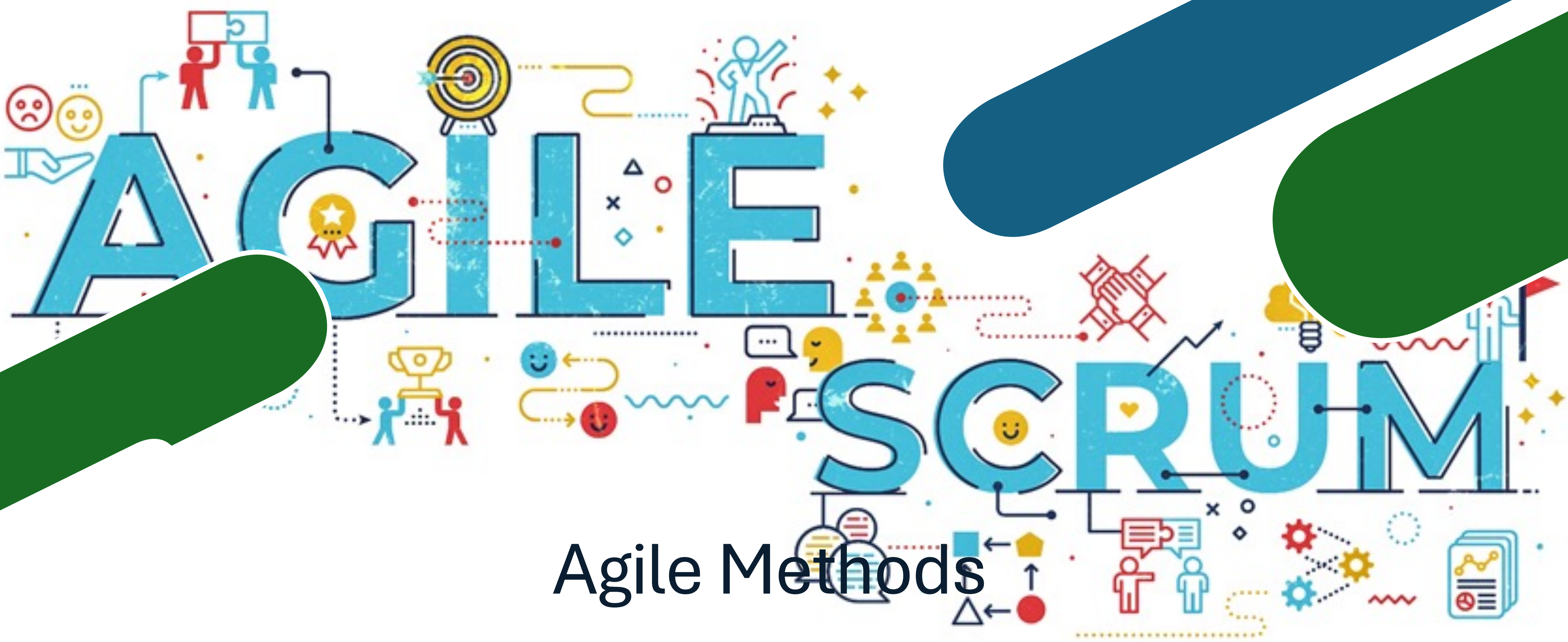
This approach, they believe, reinforces the agile strategy: simple, rapid, flexible, and user-oriented.



Agile Approaches

The Agile methodologies outlined below share much of the same overarching philosophy, as well as many of the same characteristics and practices.

- **Scrum**
- **Lean**
- **Kanban**
- **XP Extreme programming**
- **Crystal**
- **Dynamic Systems Development Method (DSDM)**
- **Feature Driven Development (FDD)**



Agile Methods

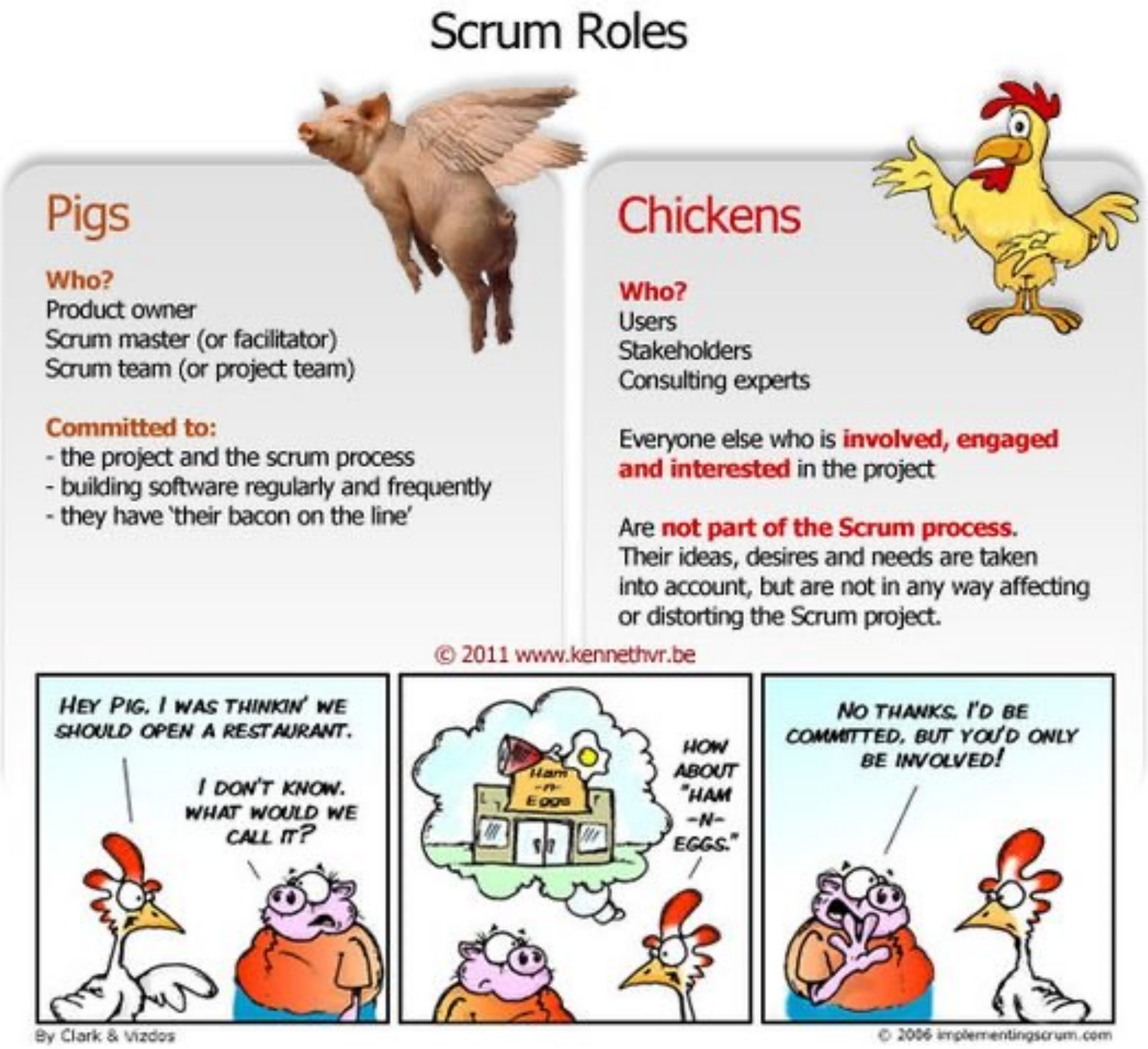
Teams can get started quickly and learn as they go. All of this makes Scrum a great choice for teams just starting to implement Agile principles.

SCRUM

The name comes from the rugby term scrum, where team members lunge at each other to achieve their objectives.

Scrum teams commit to ship working software through set intervals called sprints.

Their goal is to create learning loops to quickly gather and integrate customer feedback.



SCRUM

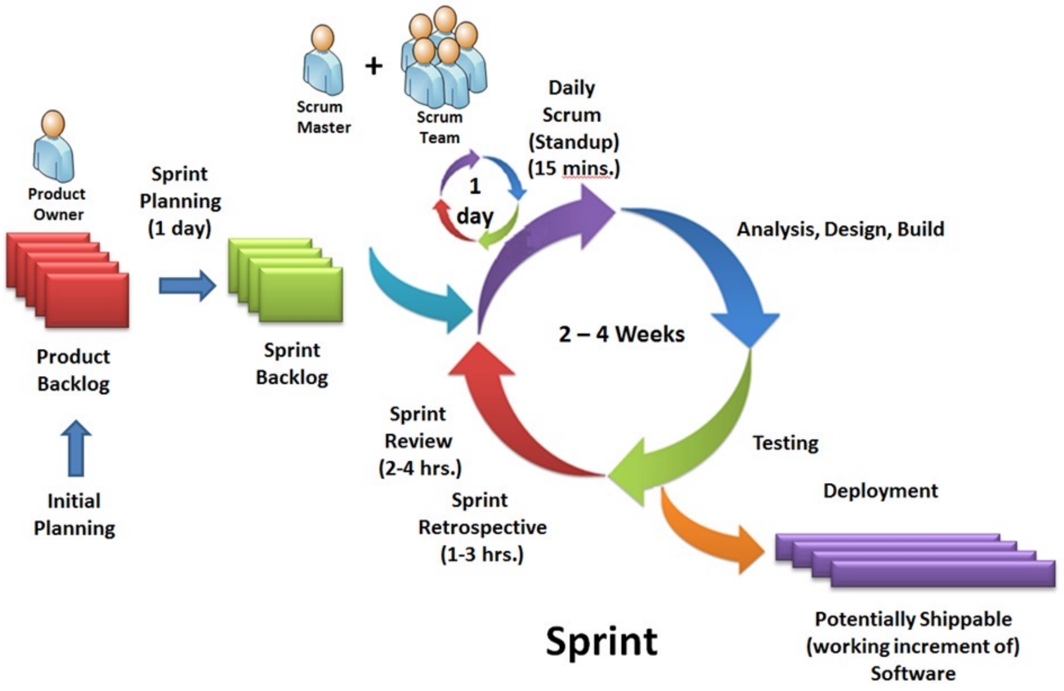
Including colorful designations such as pigs or chickens. These roles are based on the old joke about the pig and chicken who discuss a restaurant where ham and eggs would be served.

An agile team uses a series of scrums to pause the action and allow the players to reset the game plan, which remains in effect until the next scrum.

Criteria To Scrum

- Scrum moves **fast**, with sprints of **two to at most four weeks** with clear start and finish dates. The short time frame forces complex tasks to be split into smaller stories, and helps the team learn quickly.

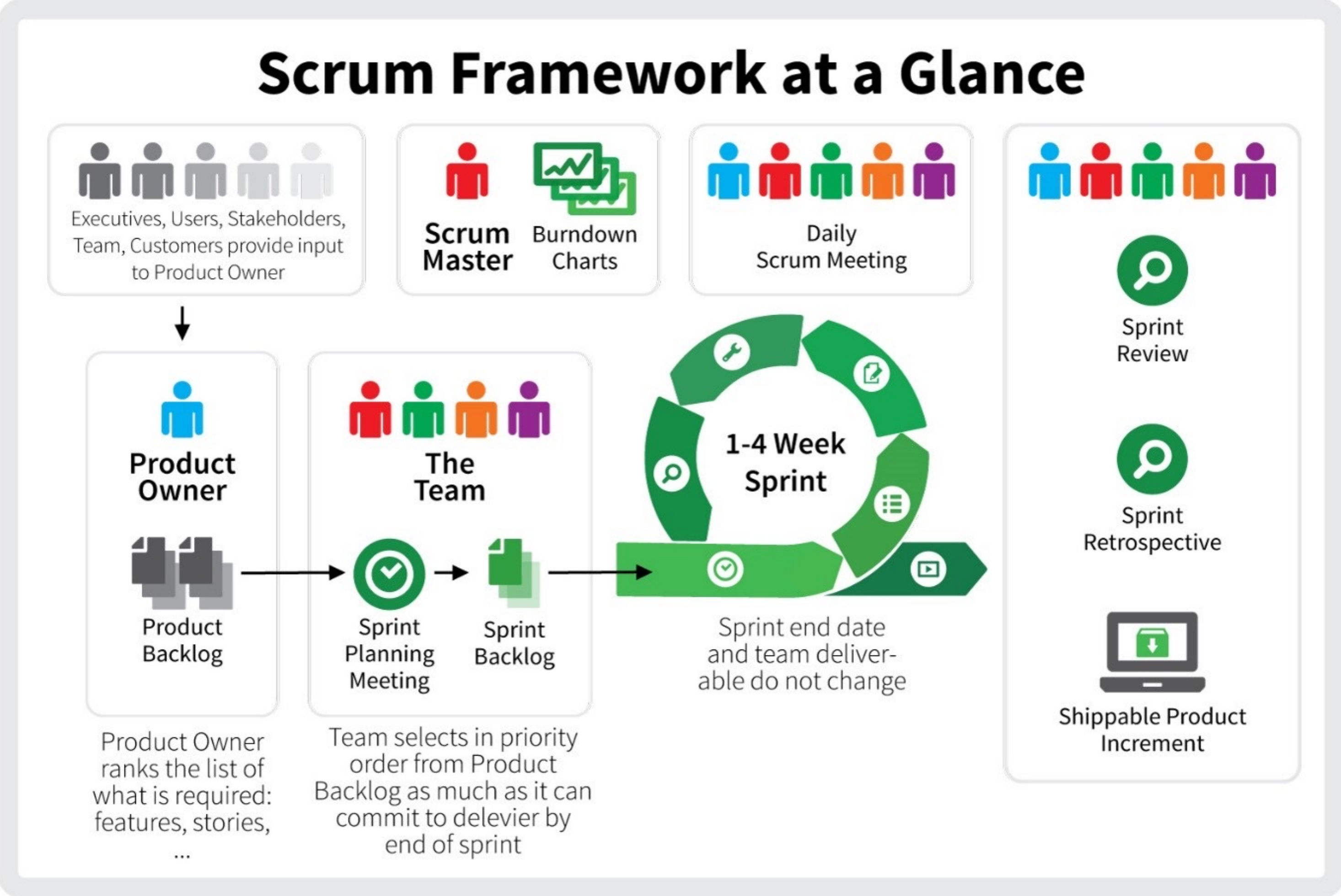
- Is Your team can be a SCRUM?
- Ask Another Question?
- key question is this:
- Can your team ship useable code that fast??????????



Scrum Rules

- Scrum has three clearly defined roles.

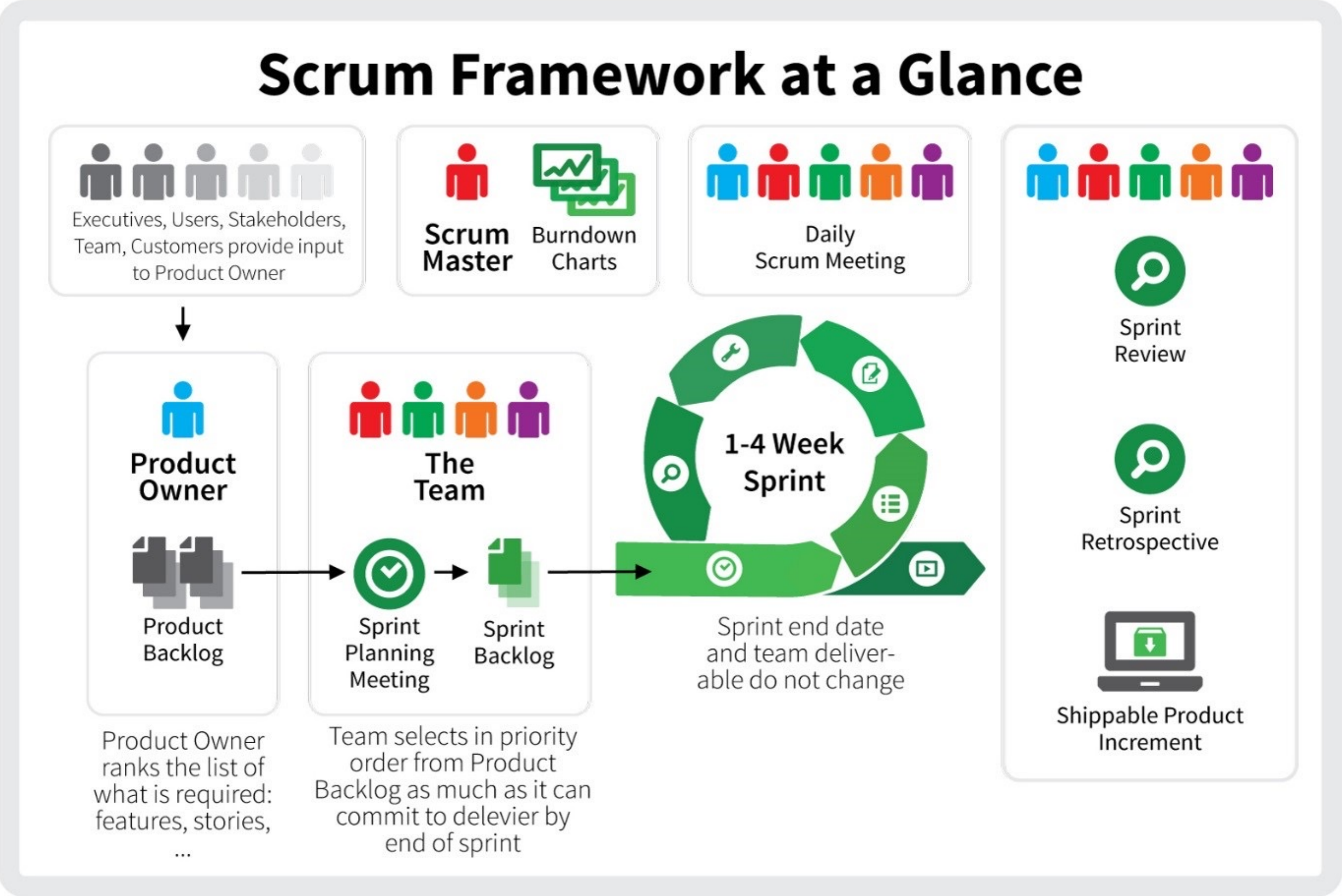
- ➢ The **product owner** advocates for the customer, manages the product backlog, and helps prioritize the work done by the development team.
- ➢ The **scrum master** helps the team stay grounded in the scrum principles.
- ➢ The **development team** chooses the work to be done, delivers increments, and demonstrates collective accountability.



WHO MANAGES THE SCRUM TEAM?

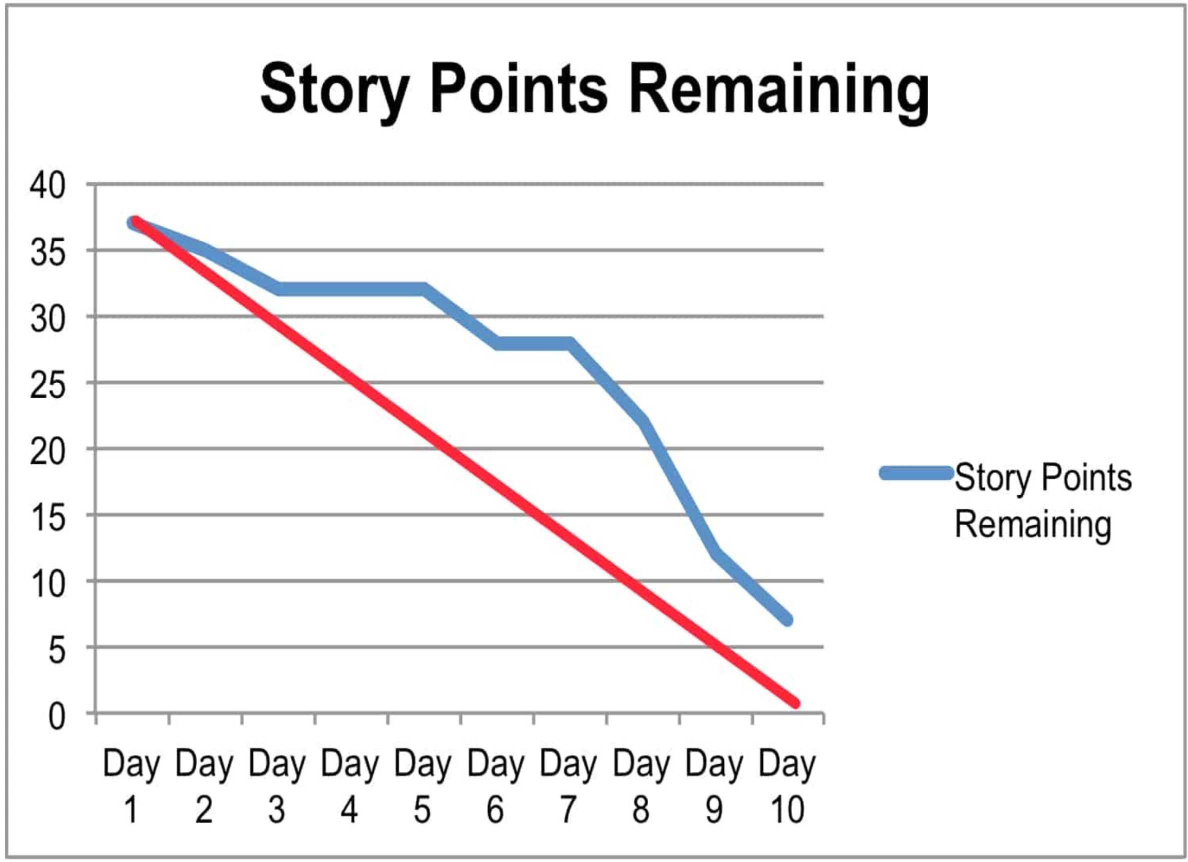
- Nobody.

- scrum teams are self-organizing and everyone is equal, despite having different responsibilities. The team is united by the goal of shipping value to customers.



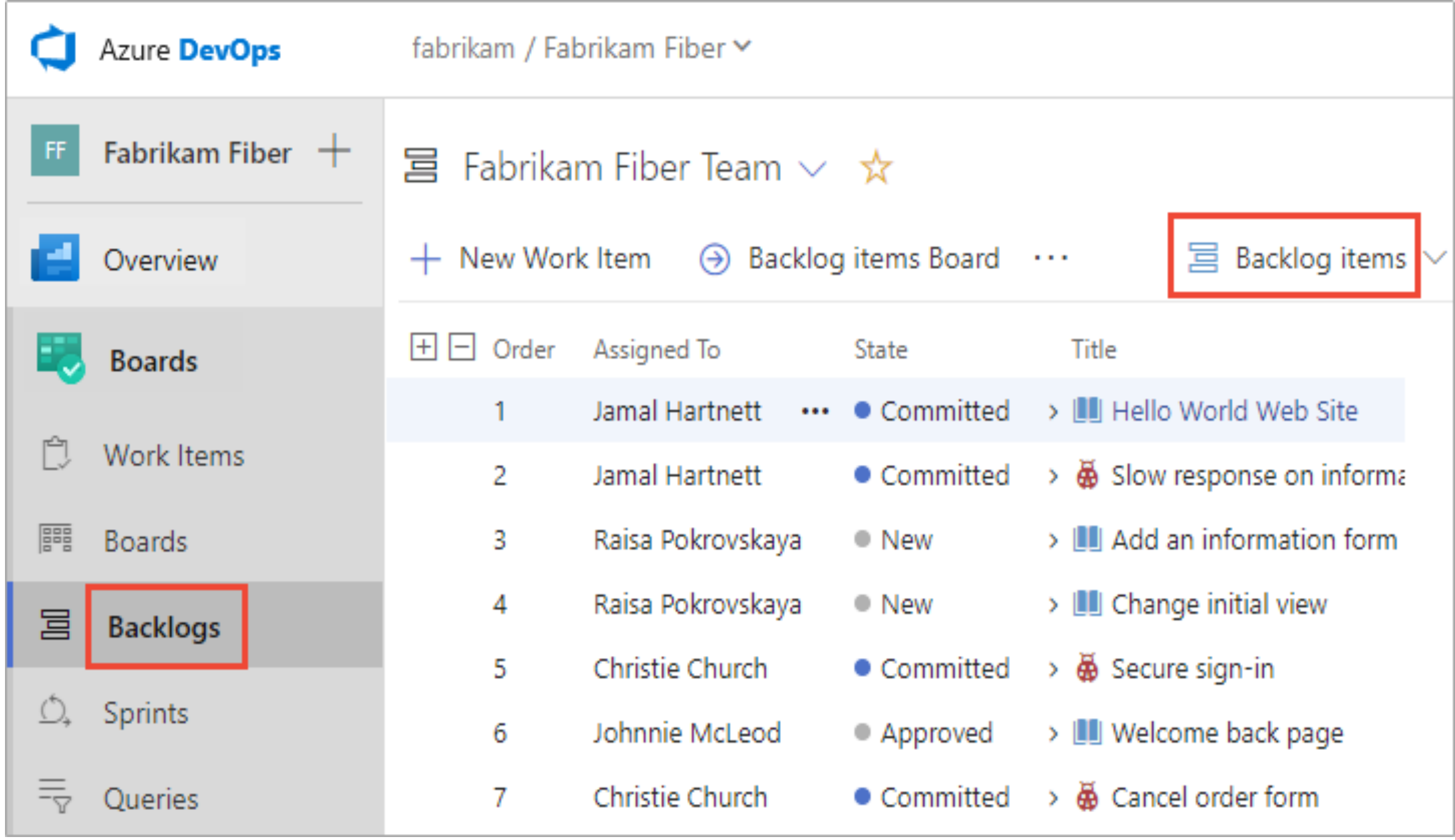
Key Metrices

- **Velocity**—the number of **story points** completed in a sprint—is the central metric for scrum teams.
- If the team completes an average of 35 story points per sprint (Velocity = 35), it won't agree to a sprint backlog that contains 45 points.
- The Burndown chart is the way to figure the velocity.

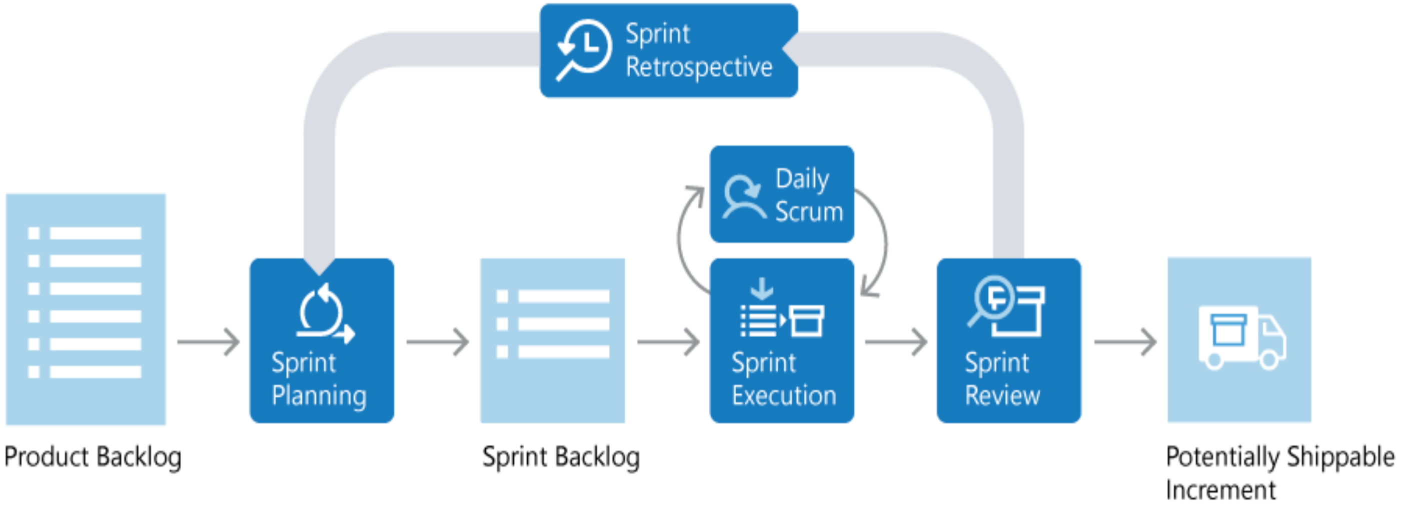


Backlogs

- A Backlog is a set of work items ordered and stated, something to do like a requirement to implement, bug to solve
- A well-prioritized agile backlog not only makes release and iteration planning easier, it broadcasts all the things the team intends to spend time on—including internal work that the customer will never notice.



Framework



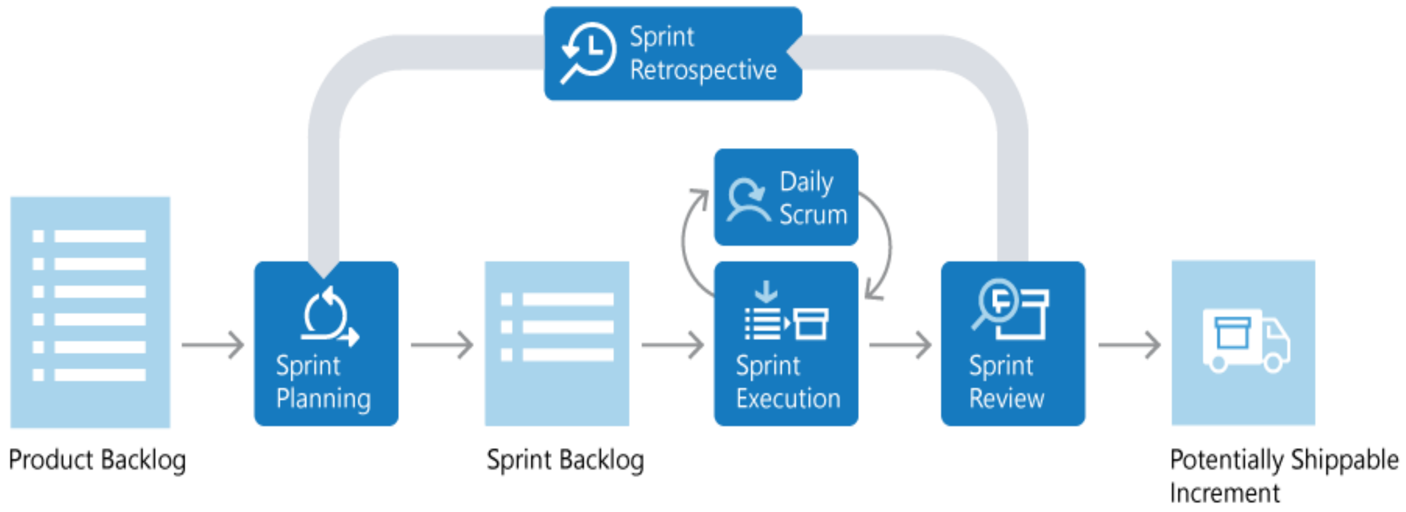
- Scrum is **iterative**. The entire lifecycle is completed in **fixed time-period** called a Sprint.
- The Product Backlog is a **prioritized** list of value the team can **deliver**. The Product Owner owns the backlog and adds, changes, and reprioritizes as needed.
- In Sprint Planning, the team chooses the **backlog items** they will work on in the upcoming sprint.

Framework

- Often, each item on the Sprint Backlog is broken down into tasks. Once all members agree the Sprint Backlog is achievable, the Sprint starts.

- Once the Sprint starts, the team executes on the Sprint Backlog. Scrum does not specify how the team should execute. That is left for the team to decide.

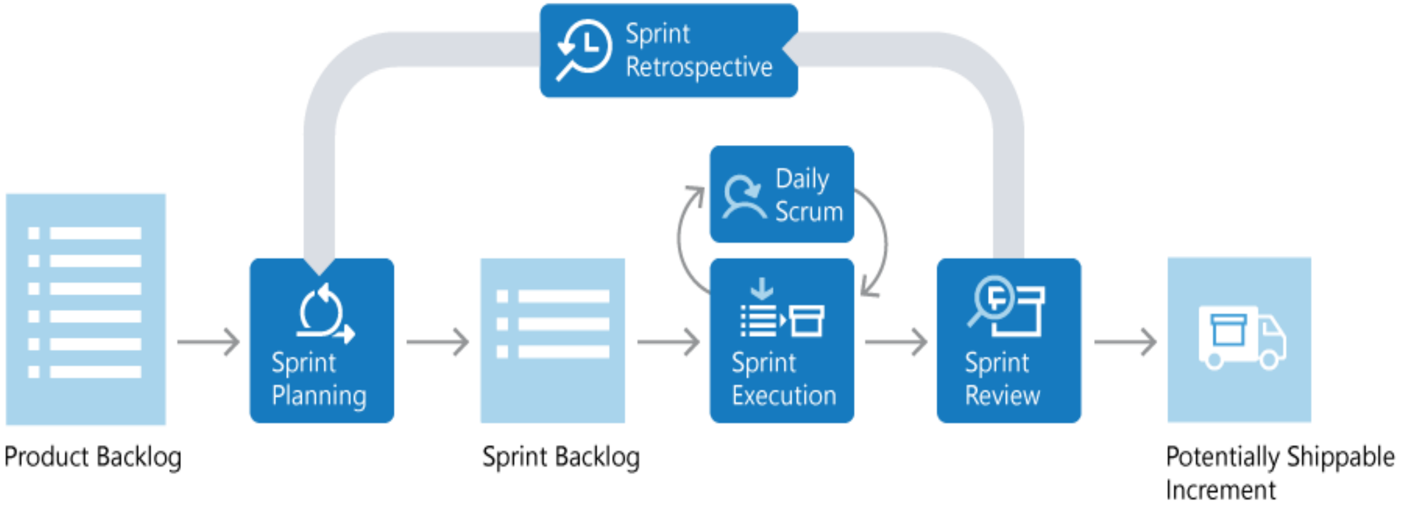
- Scrum defines a practice called a Daily Scrum, often called the Daily Standup. The Daily Scrum is daily meeting limited to 15 minutes.



Framework

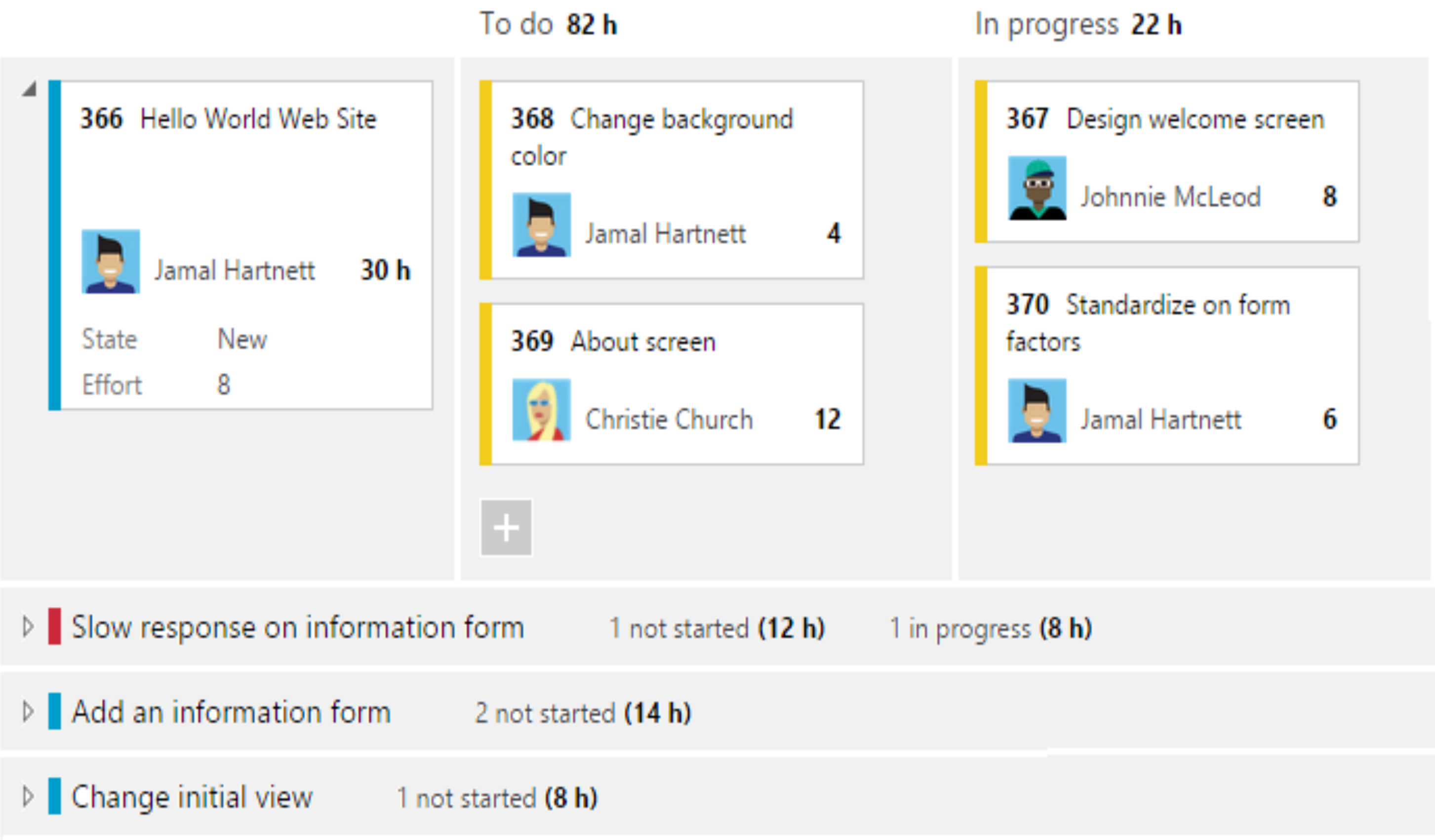
- To aid the Daily Scrum, teams often review two artifacts:

- **THE TASK BOARD**
- THE SPRINT BURNDOWN



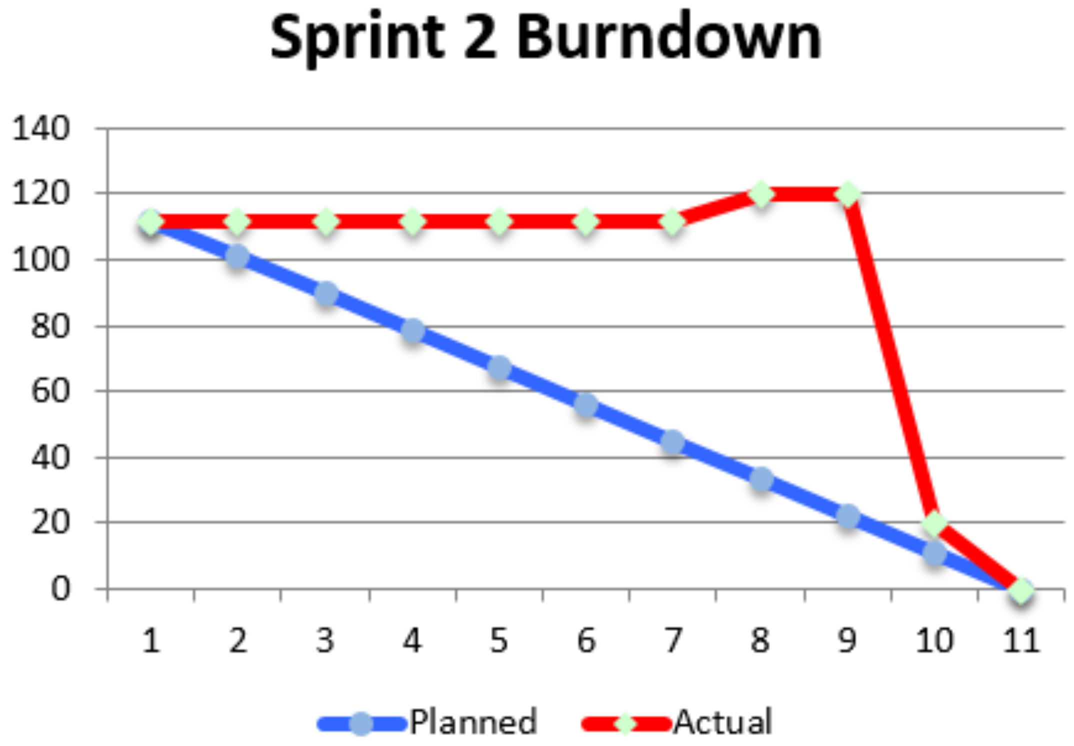
Task Board

- Lists each backlog item the team is working on, broken down into the tasks required to complete it.
- Tasks are placed in To Do, In Progress, and Done columns based on their status.
- It provides a visual way of tracking progress for each backlog item



THE SPRINT BURNDOWN

- A graph that plots the daily total of remaining work. Remaining work is typically in hours. It provides a visual way of showing whether the team is "on track" to complete all the work by the end of the Sprint.



- Sprint Review: The team demonstrates what they've accomplished to stakeholders.
- Sprint Retrospective: The team takes time to reflect on what went well and which areas need improvement.

INCREMENT

- The product of a Sprint is called the "Increment" or "Potentially Shippable Increment".
- Regardless the term, a Sprint's output should be of shippable quality, even if it's part of something bigger and can't ship by itself.
- It should meet all the quality criteria set by the team and Product Owner.

REPEAT. LEARN. IMPROVE.

- The entire cycle is repeated for the next sprint.
- Sprint Planning selects the next items on the Product Backlog and the cycle repeats.
- This shorter, iterative cycle provides the team with lots of opportunities to learn and improve.
- Scrum is very popular because it provides just enough framework to guide teams, while giving them flexibility in how they execute.

AGILE METHOD ADVANTAGES AND DISADVANTAGES

- (+) Agile, or adaptive, methods are very flexible and efficient in dealing with change.
- (+) Frequent deliverables constantly validate the project and reduce risk.
- (+) stress team interaction and reflect a set of community based values.
- (-) Team members need a high level of technical and interpersonal skills.
- (-) A lack of structure and documentation can introduce risk factors, such as blurring of roles and responsibilities, and loss of corporate knowledge.

AGILE, SCRUM AND KANBAN

While broadly fitting under the umbrella of Agile, both Scrum and Kanban are quite different.

SCRUM

- focuses on fixed length sprints.
- Scrum has defined roles
- Scrum uses velocity as a key metric.
- My Advice is to use it when IT team support many organizations like IT company.

Kanban

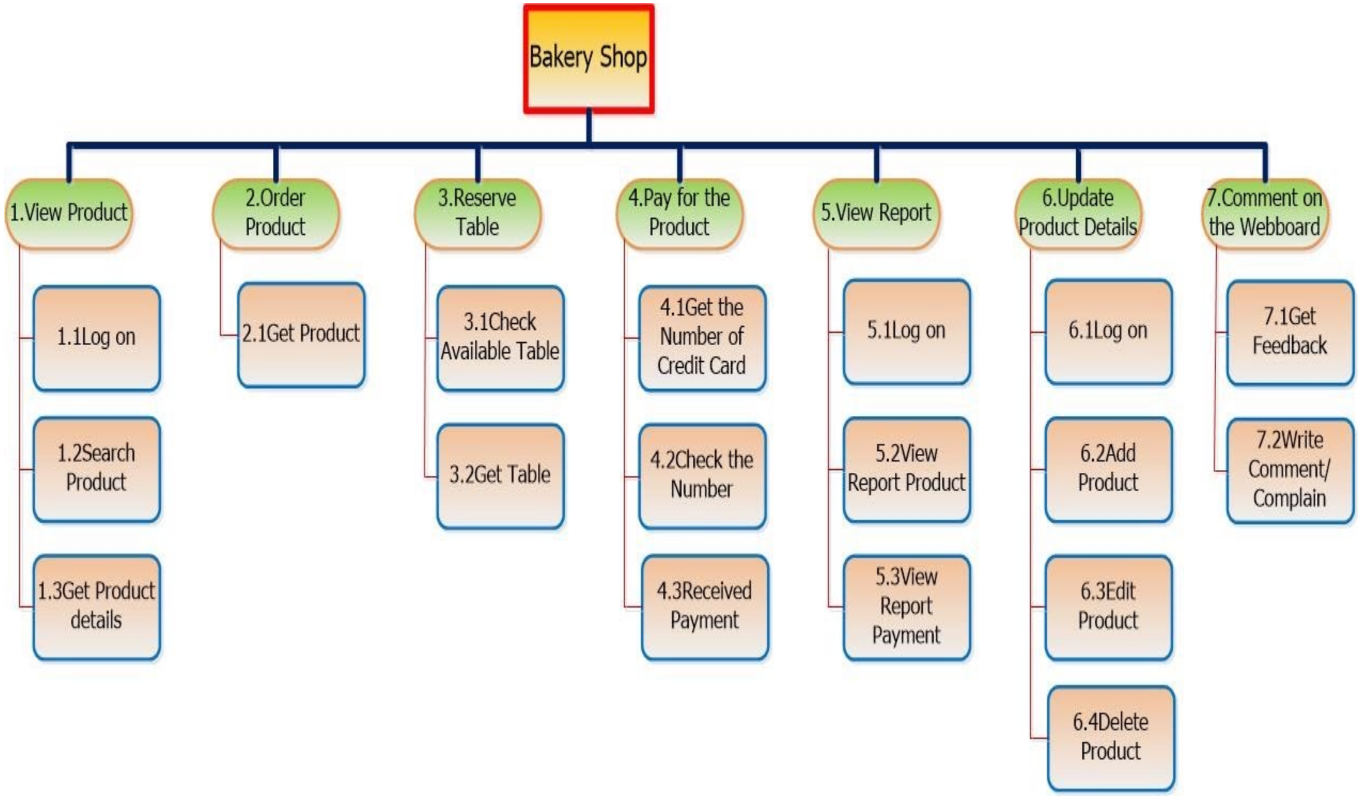
- Kanban is more of a continuous flow model.
- Kanban does not define any specific roles for the team.
- Kanban champions the use of cycle time.
- My Advice is to use it when IT team support one organization.

Modeling Techniques

Models help users, managers, and IT professionals understand the design of a system. Modeling involves graphical methods and nontechnical language that represent the system at various stages of development.

FUNCTIONAL DECOMPOSITION DIAGRAMS

- A functional decomposition diagram (FDD) is a **top-down representation** of a function or process.
- Creating an FDD is similar to drawing an **organization chart**: Start at the top and work downwards.

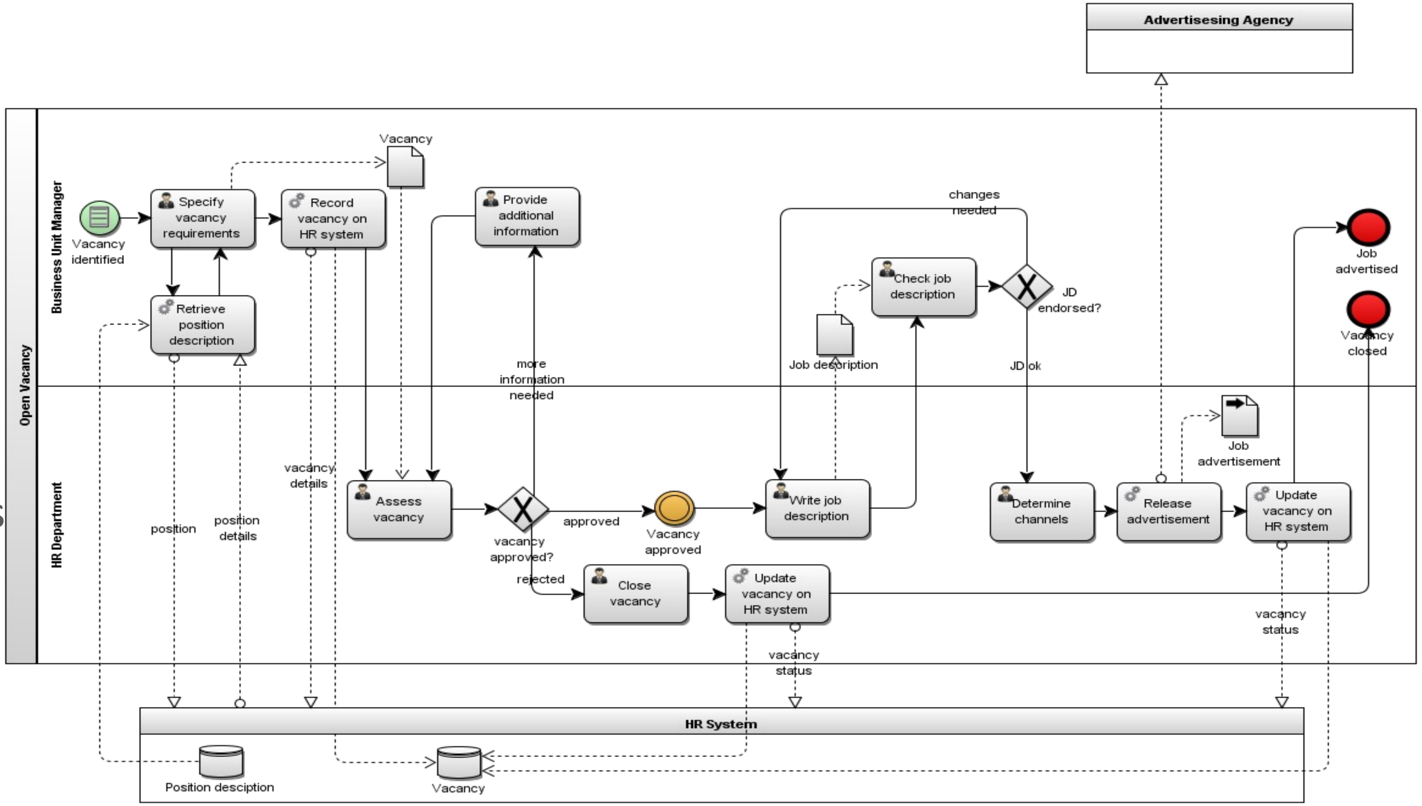


BUSINESS PROCESS MODELING

- A business process model (BPM) represents one or more **business processes.**

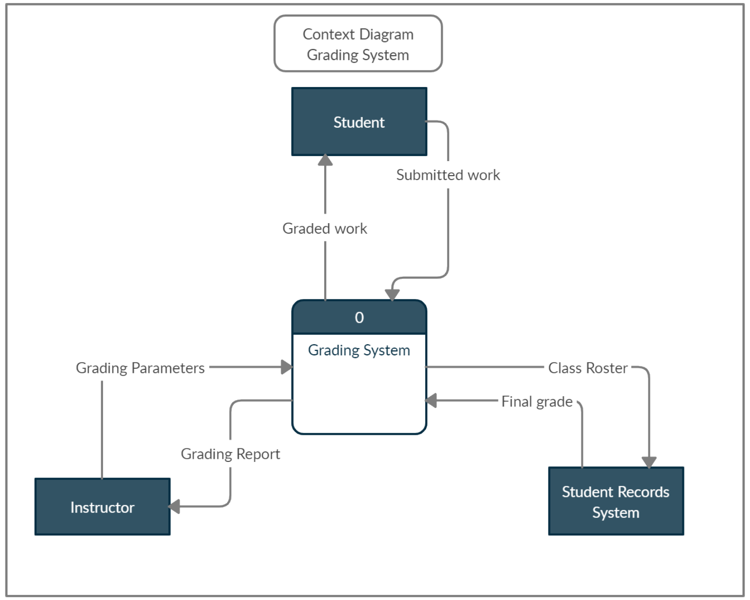
- During requirements modeling, analysts often create models that use a standard language called **business process modeling notation (BPMN).**

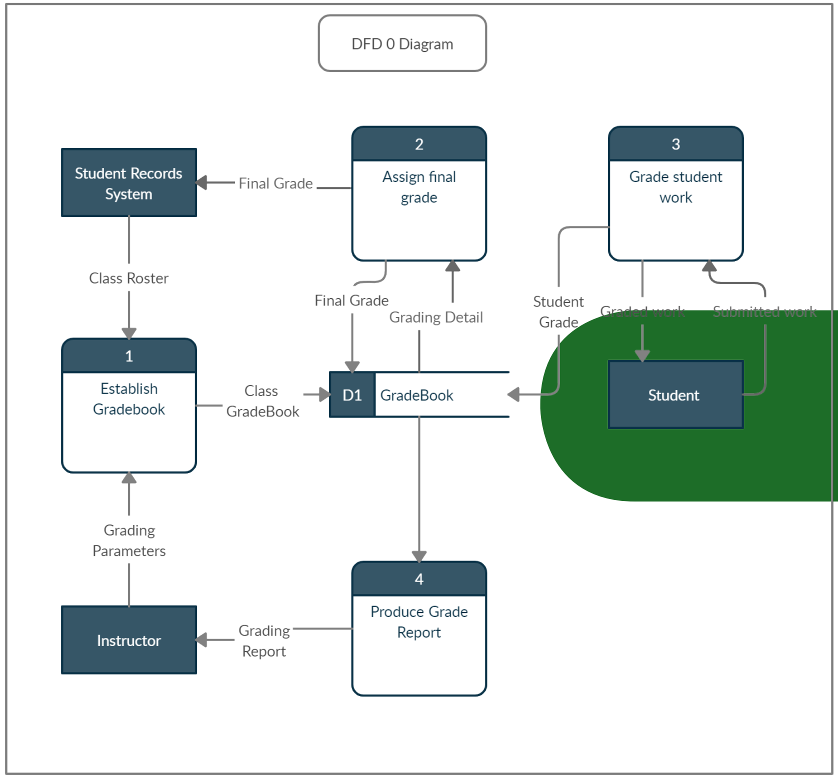
- BPMN includes various shapes and symbols to represent events, processes, and workflows.



DATA FLOW DIAGRAMS

- Working from a functional decomposition diagram, analysts can create data flow diagrams (DFDs) to show how the system stores, processes, and transforms data.





UNIFIED MODELING LANGUAGE

- **The Unified Modeling Language (UML)** is a widely used method of visualizing and documenting software systems design.
- UML uses object-oriented design concepts, but it is independent of any specific programming language and can be used to describe business processes and requirements generally.
- Use Case Diagram
- Sequence Diagram

THANK YOU!